

TAIS Project: Let's Tackle Sycophancy from the Source & Where It Matters Most



Sycophancy /ˈsɪk.ə.fən.si/ noun
the continued tendency of models to tailor their responses to the user's perceived beliefs and preferences, sometimes even at the cost of accurate, truthful information



Sycophancy as a Safety Concern

Sycophancy has been studied as a capability glitch: a sycophantic model is a model truth-agnostic and thus less capable of consistent performance. However, recent research also suggests that sycophancy is a safety issue: sycophantic models (a) are **perceived as untrustworthy agents**, (b) **propagate misinformation**, and (c) **inhibit pro-social behaviour** in humans, fostering a destabilised society of distorted motivations and volatile outcomes.



Problem

While sycophancy could be amplified by reward-hacking during RLHF, it is likely learnt from biases in training data. What remains unknown are features that trigger it. Previous research suggests variables such as: statement objectivity, first-person statements, and context domain. Unlike the other variables, context domain has not yet been investigated in isolation, despite it having the most potential to narrow down the search for relevant biases in training data and help map areas of society most exposed to risk from sycophantic behaviour.



Project Goals

This project conducts a comprehensive analysis of sycophancy as a function of context domain, including:

1. making **novel observations** on how sycophancy levels vary across context domains
2. **replicating the effect of statement objectivity** while controlling for context domain
3. linking **context domain distribution in RLHF** to observed sycophancy patterns
4. **developing hypotheses** on bias in training data causing sycophancy based on patterns across context domains



Agenda

- ✓ label open-source preferential RLHF data for context domain (using LLM-generated labels + clustering)
- ✓ compose a dataset of objective and subjective tasks for each context domain
- ✓ evaluate sycophancy on objective tasks, using LLM judge via UK AISI Inspect
- ✓ evaluate sycophancy on subjective tasks as a baseline, using LLM judge via UK AISI Inspect
- verify judge fit for purpose via a controlled experiment with human participants
- compare sycophancy levels across context domains to (a) each other, (b) RLHF data, (c) subjectivity baseline



Impact

Practical benefits of this research include:

1. narrowing down the search for **specific biases** in training data that foster sycophancy
2. providing empirical data **specifying domains** that are most at risk to manipulation of public opinion
3. feeding **targeted mitigations** as opposed to blanket measures